

=== GNB (1785495785) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth
To run the srsRAN Project gNB with this config, use the following command:

```
# sudo ./gnb -c gnb_zmq.yaml
```

cu_cp:

amf:

```
addr: 10.53.1.2          # The address or hostname of the AMF.
port: 38412
bind_addr: 10.53.1.3     # A local IP that the gNB binds to for traffic from the AMF.
supported_tracking_areas:
- tac: 7
  plmn_list:
  - plmn: "21401"
    tai_slice_support_list:
    - sst: 1
```

```
inactivity_timer: 7200    # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200]
```

ru_sdr:

```
device_driver: zmq        # The RF driver name.
device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass argument
srate: 11.52              # RF sample rate might need to be adjusted according to selected bandwidth.
tx_gain: 75               # Transmit gain of the RF might need to adjusted to the given situation.
rx_gain: 75               # Receive gain of the RF might need to adjusted to the given situation.
```

cell_cfg:

```
dl_arfcn: 630000          # ARFCN of the downlink carrier (center frequency).
band: 78                  # The NR band.
channel_bandwidth_MHz: 20 # Bandwidth in MHz. Number of PRBs will be automatically derived.
common_scs: 15            # Subcarrier spacing in kHz used for data.
plmn: "21401"             # PLMN broadcasted by the gNB.
tac: 7                    # Tracking area code (needs to match the core configuration).
pdcch:
  common:
    ss0_index: 0          # Set search space zero index to match srsUE capabilities
    coresets0_index: 6    # Set search CORESET Zero index to match srsUE capabilities
  dedicated:
    ss2_type: common      # Search Space type, has to be set to common
    dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
prach:
  prach_config_index: 1   # Sets PRACH config to match what is expected by srsUE
pdsch:
  mcs_table: qam64        # Sets PDSCH MCS to 64 QAM
pusch:
  mcs_table: qam64        # Sets PUSCH MCS to 64 QAM
```

log:

```
filename: /tmp/gnb.log    # Path of the log file.
all_level: info           # Logging level applied to all layers.
hex_max_size: 0
```

pcap:

```
mac_enable: false        # Set to true to enable MAC-layer PCAPs.
mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
ngap_enable: false        # Set to true to enable NGAP PCAPs.
ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
e2ap_enable: true         # Set to true to enable E2AP PCAPs.
e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.
```

#e2:

```
# enable_du_e2: true      # Enable DU E2 agent (one for each DU instance)
# e2sm_kpm_enabled: true  # Enable KPM service module
```

```
# e2sm_rc_enabled: true          # Enable RC service module
# addr: 127.0.0.1                # RIC IP address
#bind_addr: 127.0.0.100         # A local IP that the E2 agent binds to for traffic from the RIC. ONLY required if
# port: 36421                    # RIC port

metrics:
  layers:
    enable_rlc: true            # Enable RLC metrics
    enable_sched: true          # Enable DU scheduler metrics
  periodicity:
    du_report_period: 1000      # Set DU statistics report period to 1s
```

=== UE (1785495785) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78
nof_carriers = 1

max_nof_prb = 106
nof_prb = 106
dl_nr_arfcn = 630000

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = deeff112233445566778899aabbccdd
imsi = 214010000000020
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1785495785) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
    - SUBSCRIBER_DB=214010000000020, deeff112233445566778899aabbccdd, opc, 63BFA50EE6523365FF14C1F45F88737D, 8000, 9, 10.4
    - NETWORK_NAME_FULL=srsRAN
    - NETWORK_NAME_SHORT=srsRAN
    - TZ=Europe/Madrid
  privileged: true
  ports:
    - "9898:9999/tcp"
  # Uncomment port to use the 5gc from outside the docker network
  # - "38412:38412/sctp"
  # - "2152:2152/udp"
  command: 5gc -c open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: ${OPEN5GS_IP:-10.53.1.2}

gnb:
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
```

```

# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's the easiest
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    file: gnb_zmq.yaml # Path to your desired config file
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2}
          bind_addr: 0.0.0.0
      metrics:
        autostart_stdout_metrics: true
        enable_json: true
      remote_control:
        bind_addr: 0.0.0.0
        enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```

=== NOTAS (1785495785) ===

- PLMN (21401) y IMSI (214010000000020) ajustados en gNB, UE y SUBSCRIBER_DB (Regla 1, 7a).
- Banda 78 y ancho de banda 20 MHz configurados en gNB y UE. dl_arfcn 630000 seleccionado para Banda 78 (Regla 2).
- K y OPC del suscriptor configurados en UE y SUBSCRIBER_DB, manteniendo el OPC de la plantilla (Regla 7b).
- bind_addr del gNB configurado a 10.53.1.3 en el YAML y 0.0.0.0 en docker-compose (Regla 5).
- Bloque ru_sdr copiado exactamente de la plantilla (Regla 6).
- Puertos ZMQ cruzados verificados entre gNB y UE (Regla 4).